

National Institute of Technology Kurukshetra

B. Tech. 1st Sessional Examination, 2026

Branch	:	R&A	Course Code	:	RAPC-210
Semester	:	4 th	Time	:	50 minutes
Course Name	:	Actuators Drives and Sensors	Max. Marks	:	15
Time	:	04:00-04:50 PM	Date	:	21/02/2026

NOTE: Write answers clearly and to the point. Draw neat sketches wherever required.

Q1 A washing machine operates using a control unit that receives inputs from a clock and user program. It controls a pump, valve, heater, and motor. Feedback is obtained from sensors measuring water level, water temperature, drum speed, and door status. Design a closed-loop control strategy for the washing cycle. Clearly identify the controlled variables, feedback signals, correction elements, and explain how the control unit coordinates sequential operations to ensure safe and stable process. [5]

Q2 A level sensor is used in a closed-loop tank control system to maintain liquid level at a desired set point. During operation, the following observations are made: The sensor output does not return to zero when the tank is emptied. Small changes in level are not detected below 2 mm. The output changes slowly after 30 minutes of continuous use. The response to sudden level change is delayed by 3 seconds. The same level gives slightly different outputs during filling and emptying. Identify and classify the relevant sensor performance parameters involved. [5]

Q3 Explain the following

- d) Loading Effect on Potentiometer
- e) Low-pass Filter
- f) Incremental Encoder